METRIC

A-A-59636
15 January 2003
SUPERSEDING
FF-B-171/32
30 November 1993

COMMERCIAL ITEM DESCRIPTION

# BEARING, BALL, ANNULAR, DOUBLE ROW, ANGULAR CONTACT, NON-FILLING SLOT, VERTEX OF CONTACT ANGLES OUTSIDE BEARING, DIMENSION SERIES 32

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers metric, double row, angular contact, annular ball bearings for general-purpose use. These bearings are not intended for use in critical aeronautical or critical special precision applications.

2. CLASSIFICATION. The ball bearings shall be classified by the class, sizes, cage materials, shield and seal configurations, precision tolerances, and lubricant and preservative requirements listed below:

Class 8 - dimension series 32

Size - bearing dimensions (see table I)

Cage materials (see table II)

Shield and seal configurations (see table III)

Precision tolerances (see table IV)

Lubricant and preservative requirements (see table V)

3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing boundary dimensions shall conform to the requirements specified in table I for each of the coded bearing sizes. The listed dimensions conform to the requirements specified for the listed bearing sizes from dimension series 32 in American Bearing Manufacturers Association (ABMA) Standard 20, "Radial Bearings of Ball, Cylindrical Roller,

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and Spherical Roller Types, Metric Design" (DoD adopted). The bearing size shall be as specified in the acquisition order (see 7.3(b)). For unlisted bearing sizes, the dimensional and dynamic load rating requirements should also be specified in the acquisition order.

TABLE I. Dimensional requirements (ABMA series 32).

| Bearing size code | Bore diameter (mm) | Outside diameter (mm) | Width (mm) | Chamfer r/min. (mm) | Minimum shaft shoulder diameter[1] (mm) | Dynamic load rating (minimum) (lb.) |
|---|---|---|---|---|---|---|
| 00 | 10 | 30 | 14.3 | 0.6 | 15 | 1,320 |
| 01 | 12 | 32 | 15.9 | 0.6 | 17 | 1,730 |
| 02 | 15 | 35 | 15.9 | 0.6 | 20 | 1,970 |
| 03 | 17 | 40 | 17.5 | 0.6 | 22 | 2,170 |
| 04 | 20 | 47 | 20.6 | 1.0 | 26 | 3,140 |
| 05 | 25 | 52 | 20.6 | 1.0 | 31 | 3,400 |
| 06 | 30 | 62 | 23.8 | 1.0 | 36 | 4,720 |
| 07 | 35 | 72 | 27.0 | 1.1 | 42 | 6,230 |
| 08 | 40 | 80 | 30.2 | 1.1 | 47 | 7,480 |
| 09 | 45 | 85 | 30.2 | 1.1 | 52 | 7,985 |
| 10 | 50 | 90 | 30.2 | 1.1 | 57 | 7,925 |
| 11 | 55 | 100 | 33.3 | 1.5 | 63.5 | 9,350 |
| 12 | 60 | 110 | 36.5 | 1.5 | 68.5 | 11,445 |
| 13 | 65 | 120 | 38.1 | 1.5 | 73.5 | 13,325 |
| 14 | 70 | 125 | 39.7 | 1.5 | 78.5 | 14,540 |
| 15 | 75 | 130 | 41.3 | 1.5 | 84 | 15,805 |
| 16 | 80 | 140 | 44.4 | 2.0 | 90 | 17,380 |
| 17 | 85 | 150 | 49.2 | 2.0 | 95 | 20,105 |
| 18 | 90 | 160 | 52.4 | 2.0 | 100 | 21,460 |
| 19 | 95 | 170 | 55.6 | 2.1 | 107 | 26,080 |
| 20 | 100 | 180 | 60.3 | 2.1 | 112 | 29,320 |

[1]Listed for reference purposes only. Shoulder height shall be determined to provide sufficient clearance for the direct application of bearing removal force against the bearing inner ring. If the required minimum clearance is not available, an alternative non-destructive bearing removal capability shall be provided.

3.2 Materials.

3.2.1 Rings. The bearing ring material shall be chromium-alloy steel 52100 (UNS G52986) as specified in ASTM International (ASTM) A 295, "Standard Specification for High-Carbon and Roller Bearing Steel" (DoD adopted). The finished rings shall not exceed the associated billet material inclusion rating that is also specified in ASTM A 295. Ring hardness shall be at least 58 HRC as defined in ASTM E 18, "Standard Test Methods for Rockwell Hardness and



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Rockwell Superficial Hardness of Metallic Materials" (DoD adopted). The grain size of the ring material shall be determined in accordance with ASTM E 112, "Standard Methods for Determining Average Grain Size" (DoD adopted).

3.2.2 Balls. The bearing ball material shall be chromium-alloy steel 52100 (UNS G52986) as specified in ASTM A 295. The finished balls shall not exceed the associated billet material inclusion rating, which is also specified in ASTM A 295. Balls shall be through-hardened no less than 60 HRC and no more than 67 HRC as defined in ASTM E 18. The grain size of the ball material shall be determined in accordance with ASTM E 112.

3.2.3 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. Metallic cage materials shall operate from -40 to 250 ºF (-40 to 121 ºC). Non-metallic cage materials shall operate from -20 to 250 ºF (-30 to 121 ºC). Unless otherwise specified in the acquisition order, the cage material shall be one of the optional coded types listed in table II (see 7.3(c)).

TABLE II. Cage materials.

| Code[1] | Material type |
|---|---|
| J | Pressed steel |
| M | Machined bronze or brass |
| P | Molded plastic (nylon 6/6 or equal) |
| T | Machined non-metallic (phenolic) |
| Y | Pressed brass |
| X | Other (specify in acquisition order) |

[1]No significant change from FF-B-171/32.

3.2.4 Shields and seals. When used, the shield and seal material shall be as recommended by the manufacturer, unless otherwise specified in the acquisition order (see 7.3(d)). The material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The shield or seal shall remain functionally effective at a minimum temperature range of -20 to 250 °F (-30 to 121 °C) for non-metallic and -40 to 250 °F (-40 to 121 °C) for metallic materials. Unless otherwise specified in the acquisition order, the shield, seal, and snap ring configuration for the bearing shall be one of the coded options listed in table III (see 7.3(e)).

3.2.5 Snap rings. When used, the bearing snap ring and associated snap ring groove shall conform to the requirements specified in ANSI/ABMA Standard 20.



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TABLE III. Shield and seal configurations.

| Code[1] | Configuration option |
|---|---|
| A | Open |
| B | Single shield |
| C | Double shield |
| D | Single seal |
| E | Double seal |
| F | Single shield and seal |
| G | Open with snap ring |
| H | Single shield on snap ring side |
| J | Single shield opposite snap ring |
| K | Double shield with snap ring |
| L | Single seal on snap ring side |
| M | Single seal opposite snap ring |
| N | Double seal with snap ring |
| P | Single shield and seal (shield on snap ring side) |
| X | Other (specify in acquisition order) |

[1]No significant change from FF-B-171/32.

3.3 Precision tolerance. The bearing precision tolerance level shall conform to Annular Bearing Engineers Committee (ABEC) class ABEC-1, ABEC-3, ABEC-5, or ABEC-7 as defined in ABMA Standard 20. The tolerance class and associated radial internal clearance shall be one of the coded options listed in table IV (see 7.3(f)).

3.3.1 Bearing precision tolerance levels and internal clearances may be affected by the addition of shields and/or seals. Manufacturers/suppliers should be consulted for availability of desired precision tolerances when shields and/or seals are required.

3.4 Dynamic load rating. The listed ratings in table I conform to the requirements specified in ANSI/ABMA Standard 9, "Load Ratings and Fatigue Life for Ball Bearings".

3.5 Lubrication and preservation requirements. Unless otherwise specified in the acquisition order, the bearing lubricant or preservative requirement shall be as selected from the approved coded options listed in table V (see 7.3(g)). When grease fill is required, the bearing void (the airspace between the inner and outer rings of the assembled bearing) shall be 25 to 50 percent filled. Bearing preservative shall be in accordance with MIL-DTL-197, "Packaging of Bearings, Associated Parts and Subassemblies".



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TABLE IV. Precision tolerance requirements.

| Code | Tolerance class | Radial internal clearance |
|---|---|---|
| A | ABEC-1 | Group 2 (less than normal) |
| B | | Group N (normal) |
| C | | Group 3 (greater than normal) |
| D | | Group 4 (greater than group 3) |
| T | | Group 5 (greater than group 4) |
| E | ABEC-3 | Group 2 (less than normal) |
| F | | Group N (normal) |
| G | | Group 3 (greater than normal) |
| H | | Group 4 (greater than group 3) |
| U | | Group 5 (greater than group 4) |
| J | ABEC-5 | Group 2 (less than normal) |
| K | | Group N (normal) |
| L | | Group 3 (greater than normal) |
| M | | Group 4 (greater than group 3) |
| V | | Group 5 (greater than group 4) |
| N | ABEC-7 | Group 2 (less than normal) |
| P | | Group N (normal) |
| R | | Group 3 (greater than normal) |
| S | | Group 4 (greater than group 3) |
| W | | Group 5 (greater than group 4) |

TABLE V. Lubricant and preservative requirements.

| Code | Lubricant or preservative compound |
|---|---|
| B | Grease in accordance with MIL-PRF-81322 |
| E | Grease in accordance with DOD-G-24508 |
| H | Grease in accordance with MIL-PRF-23827 |
| L | Grease with SRI-2 or qualified equivalent |
| S | No fill |
| T | Preservation compound in accordance with MIL-DTL-197 |
| X | Other (specify in the acquisition order) |



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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Unless otherwise indicated in the solicitation and resulting contract, the foreign acquisition restrictions in Section 225.7019 of the Defense Federal Acquisition Regulation Supplement (DFARS) apply to products described by this CID.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

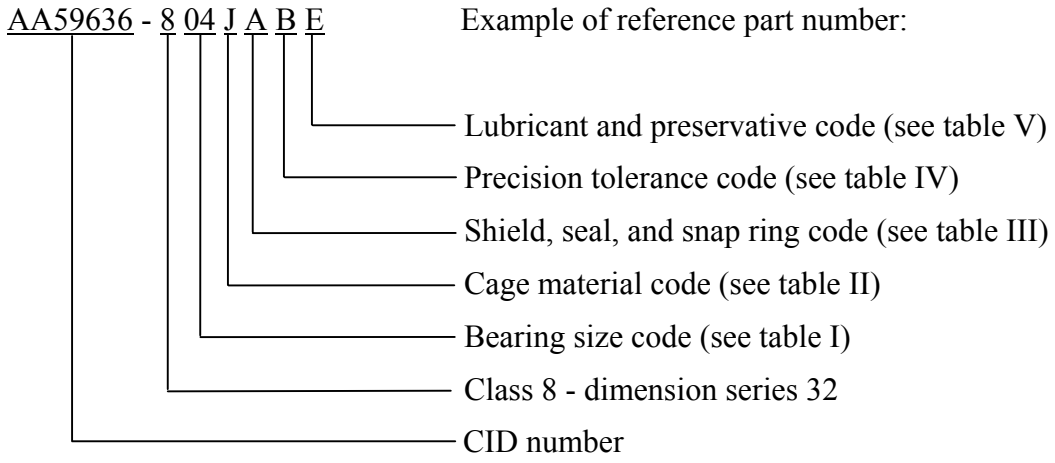
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Unless otherwise specified in the acquisition order, the bearings supplied shall be preserved, packed, and marked in accordance with MIL-DTL-197 (see 7.3(h)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.




AA59636 - 8 04 J A B E indicates: boundary dimension series 32; bore diameter 20 mm, outside diameter 47 mm, width 20.6 mm; pressed steel cage; open (no shields or seals); ABEC-1 tolerance class, normal radial internal clearance; filled with grease in accordance with DOD-G-24508.



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7.2 Sources of documents.

7.2.1 ABMA standards. Copies of ABMA standards may be obtained from the American Bearing Manufacturers Association, 2025 M Street NW, Suite 800, Washington, DC 20036. Electronic copies of ABMA standards may be obtained from http://www.abma-dc.org/.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

7.2.3 FAR and DFARS. Copies of the FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far/. Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dp/dars/dfars.html.

7.2.4 Military specifications. Copies of military specifications may be obtained from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies of military specifications may be obtained from http://astimage.daps.dla.mil/quicksearch/.

7.3 Ordering data. Acquisition documents should specify the following information:

a. CID document number, revision, and CID PIN.
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1).
c. Cage material type (see 3.2.3).
d. Shield and seal material (see 3.2.4).
e. Shield, seal, and snap ring configuration (see 3.2.4).
f. Precision tolerance requirements (see 3.3).
g. Lubricant and preservative requirement (see 3.4).
h. Preservation, packing, and marking requirements (see 6.1).

7.4 Codes cross-reference. Tables VI, VII, and VIII contain cross-reference data for the PIN information as listed in FF-B-171/32 and this CID.

7.4.1 Bearing class designations. The CIDs replacing 33 of the 37 associated specification sheets of FF-B-171 have been assigned class codes corresponding to ABMA dimension series. Table VI lists the FF-B-171 specification sheets, the corresponding dimension series, the CID class codes, and the replacement CIDs.



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TABLE VI. Federal specification to CID cross-reference.

| FF-B-171 specification sheets | ABMA dimension series | CID class codes | Replacement CID numbers |
|---|---|---|---|
| 1 | 02, 03, 10 | 1, 2, 4 | A-A-59581 |
| 2 | 19 | 5 | A-A-59582 |
| 3 | 10 | 4 | A-A-59583 |
| 4 | 02 | 1 | A-A-59584 |
| 5 | 03 | 2 | A-A-59585 |
| 6 | 04 | 3 | A-A-59586 |
| 7 | 02 | 1 | A-A-59587 |
| 8 | 03 | 2 | A-A-59589 |
| 9 | 32 | 8 | A-A-59595 |
| 10 | 33 | 9 | A-A-59596 |
| 11 | 02 | 1 | A-A-59597 |
| 12 | 03 | 2 | A-A-59598 |
| 13 | 22 | 6 | A-A-59599 |
| 14 | 23 | 7 | A-A-59600 |
| 15 | 19 | --- | Canceled |
| 16 | 10 | --- | Canceled |
| 17 | 02 | --- | Canceled |
| 18 | 03 | --- | Canceled |
| 19 | 19 | 5 | A-A-59623 |
| 20 | 10 | 4 | A-A-59624 |
| 21 | 02 | 1 | A-A-59625 |
| 22 | 03 | 2 | A-A-59626 |
| 23 | 04 | 3 | A-A-59627 |
| 24 | 10 | 4 | A-A-59628 |
| 25 | 02 | 1 | A-A-59629 |
| 26 | 03 | 2 | A-A-59630 |
| 27 | 04 | 3 | A-A-59631 |
| 28 | 10 | 4 | A-A-59632 |
| 29 | 02 | 1 | A-A-59633 |
| 30 | 03 | 2 | A-A-59634 |
| 31 | 04 | 3 | A-A-59635 |
| 32 | 32 | 8 | A-A-59636 |
| 33 | 33 | 9 | A-A-59637 |
| 34 | 32 | 8 | A-A-59638 |
| 35 | 33 | 9 | A-A-59639 |
| 36 | 32 | 8 | A-A-59640 |
| 37 | 33 | 9 | A-A-59641 |



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TABLE VII. Radial internal clearance and ABEC tolerance codes.

| FF-B-171/32 codes | | | A-A-59636 codes | | |
|---|---|---|---|---|---|
| Code | Radial internal clearance | Tolerance class | Code | Tolerance class | Radial internal clearance |
| 1 | Symbol 2 | ABEC-1 | A | ABEC-1 | Group 2 |
| 2 | Symbol 0 | | B | | Group N |
| 3 | Symbol 3 | | C | | Group 3 |
| 4 | Symbol 4 | | D | | Group 4 |
| | | | T | | Group 5 |
| | | | E | ABEC-3 | Group 2 |
| | | | F | | Group N |
| | | | G | | Group 3 |
| | | | H | | Group 4 |
| | | | U | | Group 5 |
| 5 | Symbol 2 | ABEC-5 | J | ABEC-5 | Group 2 |
| 6 | Symbol 0 | | K | | Group N |
| 7 | Symbol 3 | | L | | Group 3 |
| 8 | Symbol 4 | | M | | Group 4 |
| | | | V | | Group 5 |
| | | | N | ABEC-7 | Group 2 |
| | | | P | | Group N |
| | | | R | | Group 3 |
| | | | S | | Group 4 |
| | | | W | | Group 5 |

TABLE VIII. Lubricant and preservative requirements codes.

| FF-B-171/32 codes | | A-A-59636 codes | |
|---|---|---|---|
| Code | Lubricant or preservative | Code | Lubricant or preservative |
| A | Grease IAW MIL-PRF-81322 | B | Grease IAW MIL-PRF-81322 |
| B | Grease IAW DOD-G-24508 | E | Grease IAW DOD-G-24508 |
| C | Grease IAW MIL-PRF-23827 | H | Grease IAW MIL-PRF-23827 |
| D | Grease with SRI-2 or equivalent | L | Grease with SRI-2 or qualified equivalent |
| E | Preservation compound IAW MIL-DTL-197 | T | Preservation compound IAW MIL-DTL-197 |
| | | S | No fill |
| | | X | Other (specify in the acquisition order) |



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7.5 Subject term (key word) listing.

ABEC
cage
dynamic load rating
lubricant
precision tolerance
rings

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - OS
Air Force - 99

Review Activities:
Army - AR, EA, GL
Navy - SH
Air Force - 11, 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing Activity:
DLA - GS4

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